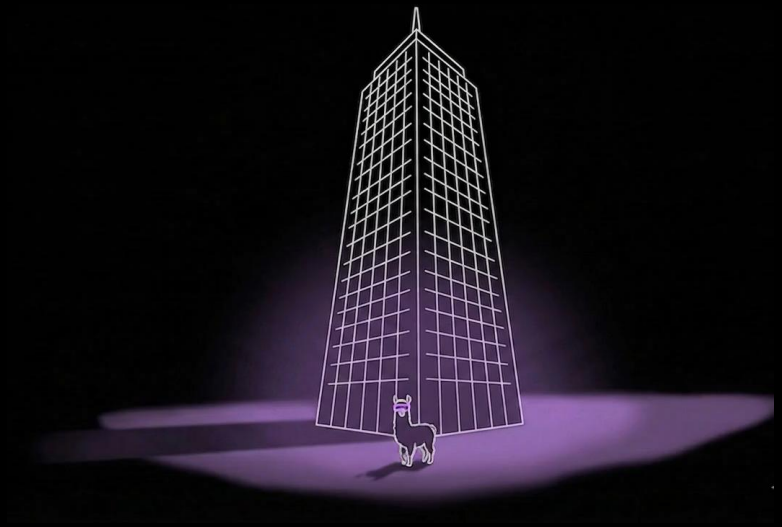


Beyond the Cloud Monolith

Building Transparent and Distributed AI with Open Source

22/04/2026

Raphaël Semeteys

WORLDLINE



Raphaël Semeteys



WORLDLINE

Head of Tech Advocacy
Senior Architect

Raphiki Yogeek
Opinions are my own

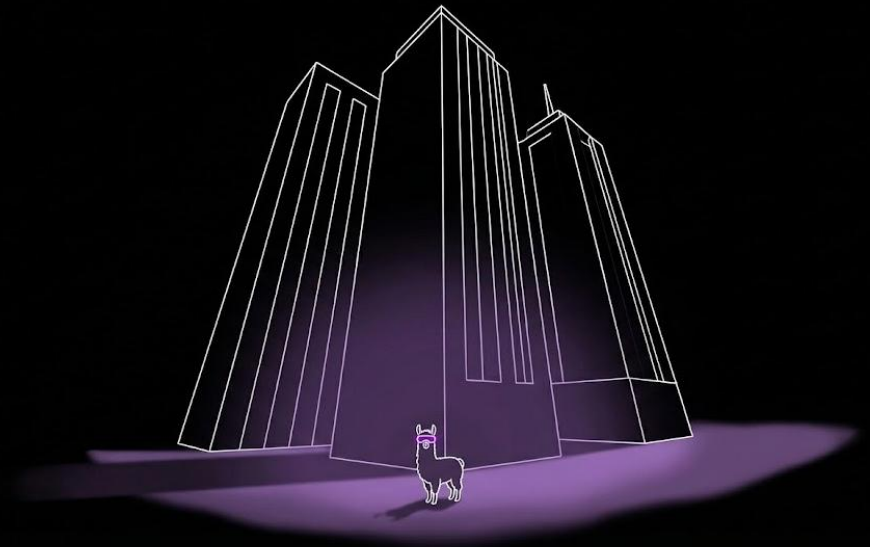
Intelligence Oligopoly

Geopolitical issues

Economic issues

Technical issues

Ecological issues





1. Defining Openness in AI




FLASH
BIACK
2024



From OpenAI to Open Source AI

Navigating between Commercial Ownership and Collaborative Openness

WORLDLINE

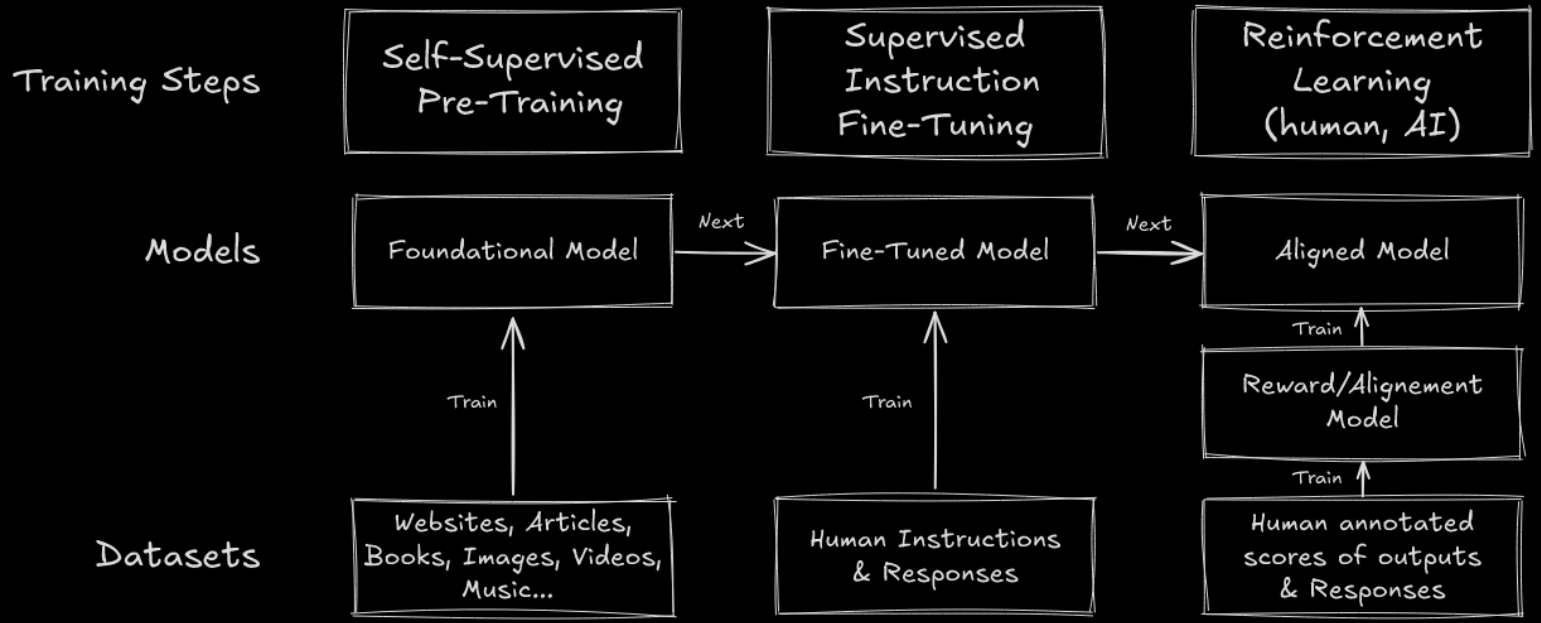
24 October 2024

OCX 24

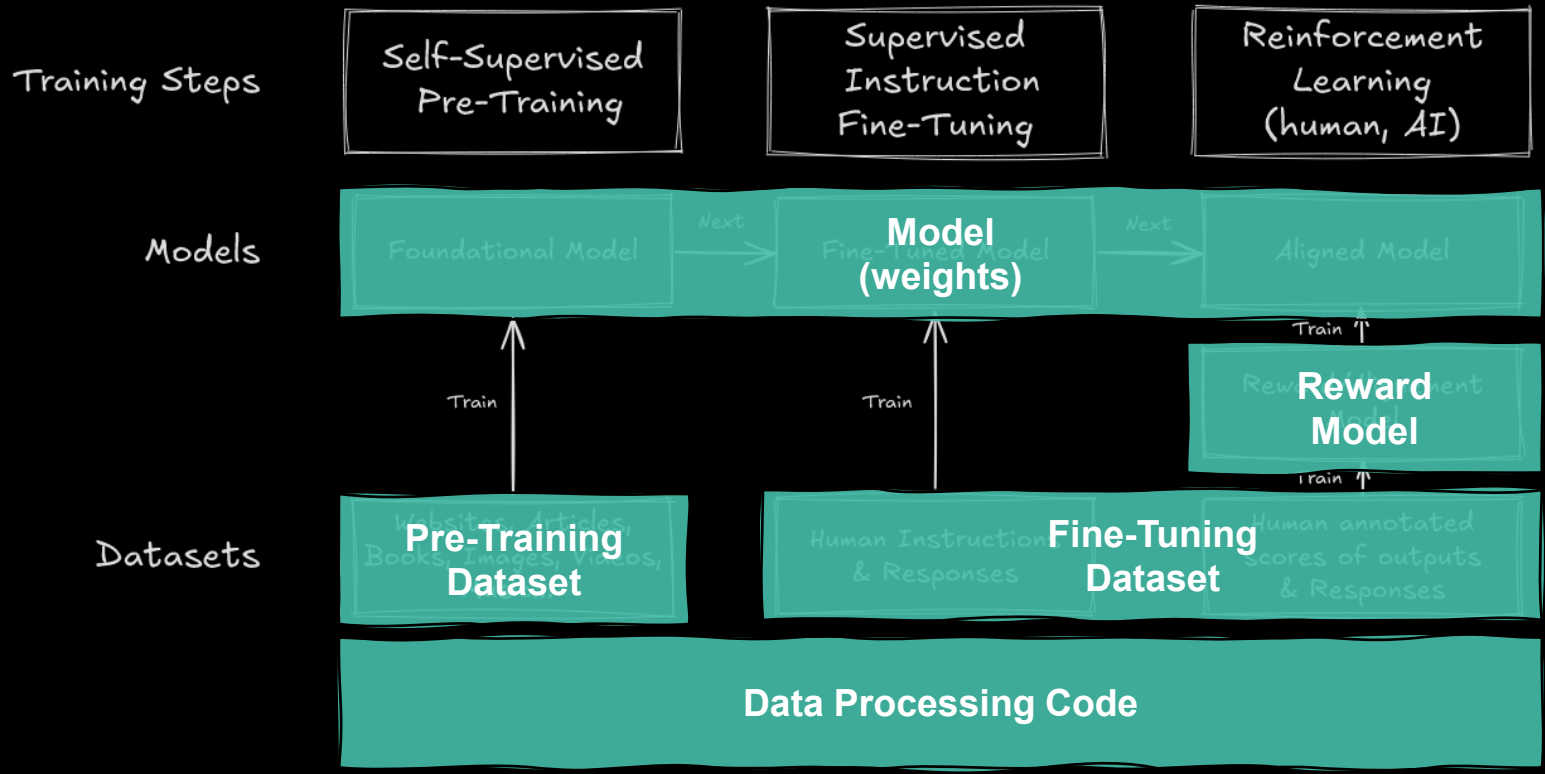
OSA OPEN COMMUNITY FOR AI

WORLDLINE

Stack #1 - AI Openness



Stack #1 - AI Openness



Open-Weight ≠ Open Source

Closed API-Only	Restricted Open-Weight	Permissive Open-Weight	Completely Open
Black Box	**Licenses restrictions**	**Private data**	**Full reproducibility**
Access only via commercial APIs	*Weights available but with constraints on use*	*Commercial freedom, undisclosed data*	*Scientific research, auditability and trust*
Gemini, GPT 3-5, Muse, Qwen3-Max, Mistral Large	LLaMA, Qwen (MAU Cap), BLOOM (RAIL License)	Gemma, GPT-OSS, Qwen, Mistral, DeepSeek, Falcon	*EleutherAI, OLMo, LLM360, Lucie*
Flagship models remain closed	License is everything	Hyperscalers Challengers	Full reproducibility (still) rare

Why Open AI

Resilience



Innovation

Trust

Openness allows Optimization & Democratization

Model Specialization (SLM)

Optimization Techniques

Existing and cheaper equipment

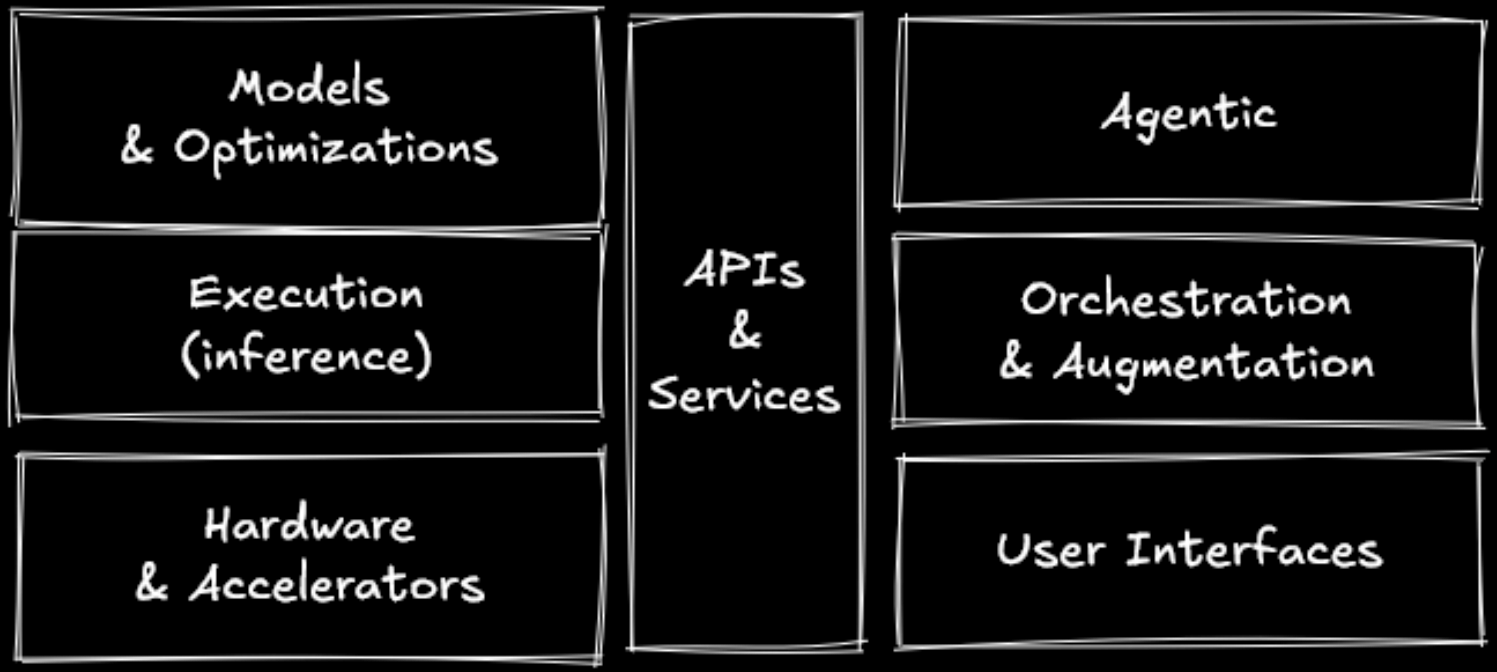
Local AI



2. Local Inference



Stack #2 - Local AI Inference



Hardware & Accelerators

General and specialized hardware

Low-level acceleration libraries

- Optimized GPU Access
- Complex compute → hardware instructions

NVIDIA CUDA

AMD ROCm

OpenVINO



Execution (inference)

Generic Frameworks

- Build, Train and Run models
- Highly flexible but not optimized for pure inference

Inference-optimized Engines

PyTorch LLaMA C++



Models & Optimizations

Open Models

Optimizations for Inference

- Pruning, Quantization
- Distillation, LoRA

ONNX & GGUF Standards

Gemma Qwen MISTRAL AI_ Phi-4



APIs & Services

Model Serving

- Performance (cache, dynamic loading)
- APIs (OpenAI-like de facto standard)
- Development (batch, stream, log, observe)

Numerous Inference Servers

Ollama vLLM Jan



User Interfaces

For End Users

- Web & Desktop UI
- Browser & Office integration

For Developers

- Notebooks
- Numerous Coding Assistants & Agents



Thunderbolt jupyter THEIA CURSOR opencode

Orchestration & Augmentation

Orchestration
- Model abstraction
- Memory Management

Retrieval Augmented Generation
- Data connectors (docs, database, web)
- Vector Stores

Integration
- Tools (APIs)
- MCP

LangChain Flowise Qdrant



Agentic AI

Agents

- Goal-oriented, planner
- Multi-agent workflows

Frameworks

- For Developers
- For End Users

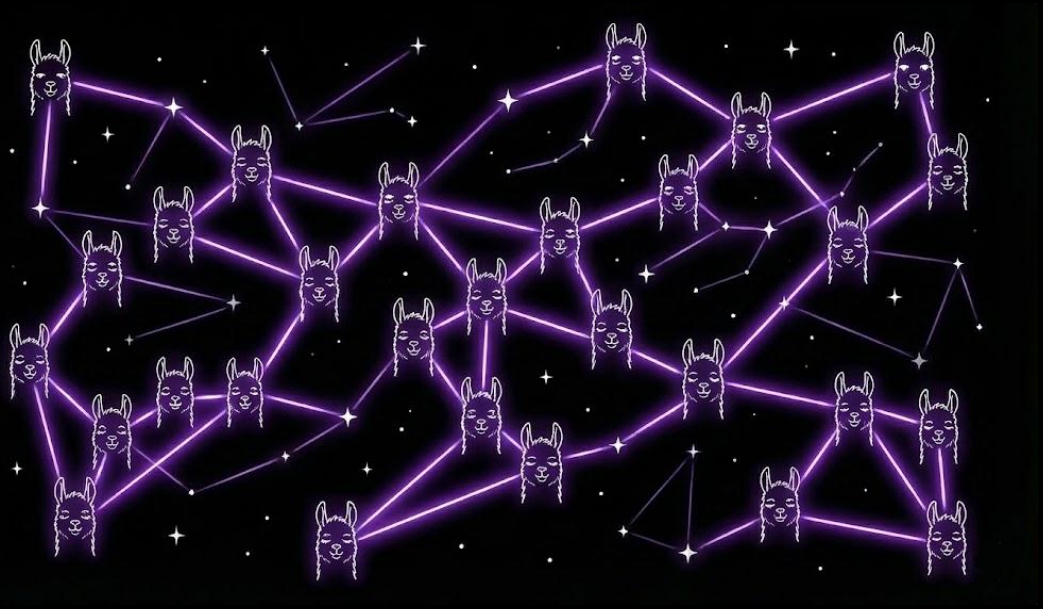
New Protocols

- Low level: MCP, A2A, TOON
- Specialized: payment (ACP, AP2)



LangGraph crewai OPENCLAW HERMES-AGENT

3. Distributed & Decentralized AI



Distributed Agentic Workflows

Open Source Frameworks

Interoperability protocols

- Model Context Protocol (MCP)
- Agent-to-Agent (A2A)

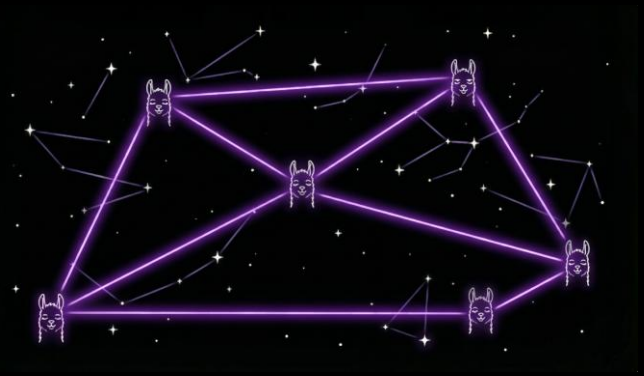


Distributed Execution

Overcoming hardware limitations

- Offloading (Local + Cloud)
- Distribution Patterns (gateway, bus)
- Inference clusters

Ollama Cloud LiteLLM llm-d RAY



Distributed AI vs Decentralized AI

Modularity

How a system is organized

Decentralization

Who controls and verifies the system

Distributed AI vs Decentralized AI

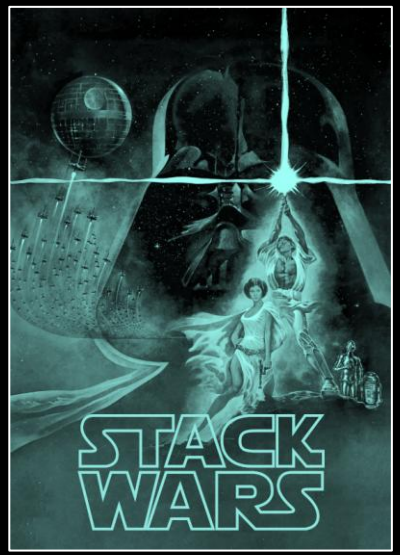
Distribution of tasks
(technical)

Distribution of power and trust
(political)


FLASH
BIACK
2023

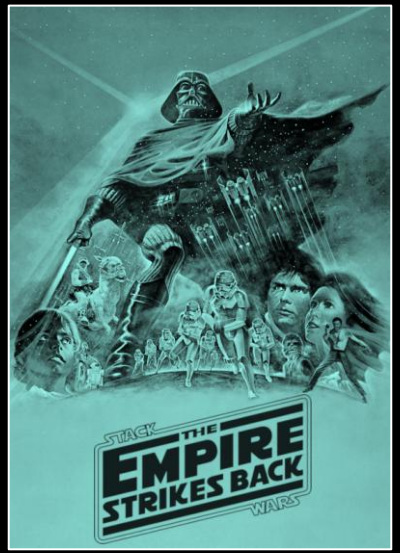
Re-decentralize the Internet

Me



Decentralization
(infrastructure)
Collective creation of the Internet

My children



Centralisation
(Business Rules/Value)
Rise of social/business platforms

Tomorrow?



Decentralization
(Business Rules/Value)
New Decentralized Protocols

Pillars of decentralization

Open to all (permissionless)

Actual Ownership

Verification (cryptographic proof)



Token (crypto not AI)

Not money
- Value Management
- Programmable tool

Key features
- Incentive
- Honesty (staking/slashing)
- Consensus



Stack #3 - Decentralized AI



Decentralized Computing

Pooling existing GPUs

How can we trust?
Why share?

io.net, Aethir

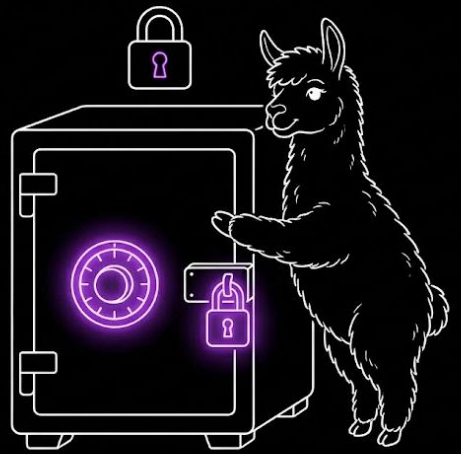


Decentralized data

Transparent, fair and proven sourcing

How can privacy be respected?
How to trace the provenance?

Grass, Vana, 0G

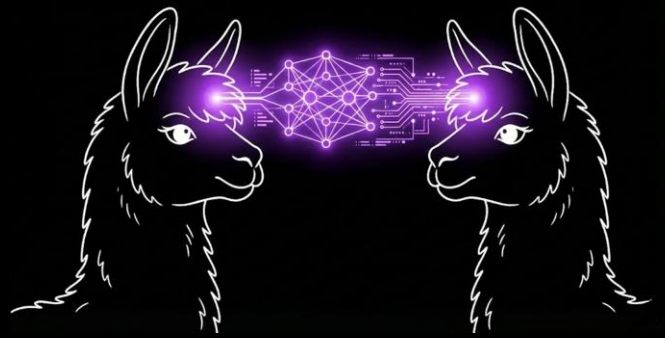


Decentralized intelligence

Improve the value created by models (inference, training)

How can we reward utility rather than calculation?

Bittensor, Prime Intellect, Pluralis



Decentralized validation

Zero knowledge (zk) Principle

How can we trust the compute but not the compute provider?

Ora, Gensys



Decentralized action

Agents as actors/autonomous economic enterprises (wallet)

How to prepare for a Machine-to-Machine economy?

Olas, Virtuals Protocol



Decentralized action

Agents as actors/autonomous economic enterprises (wallet)

How to prepare for a Machine-to-Machine economy?

Olas, Virtuals Protocol



AI + Crypto = ecological disaster?

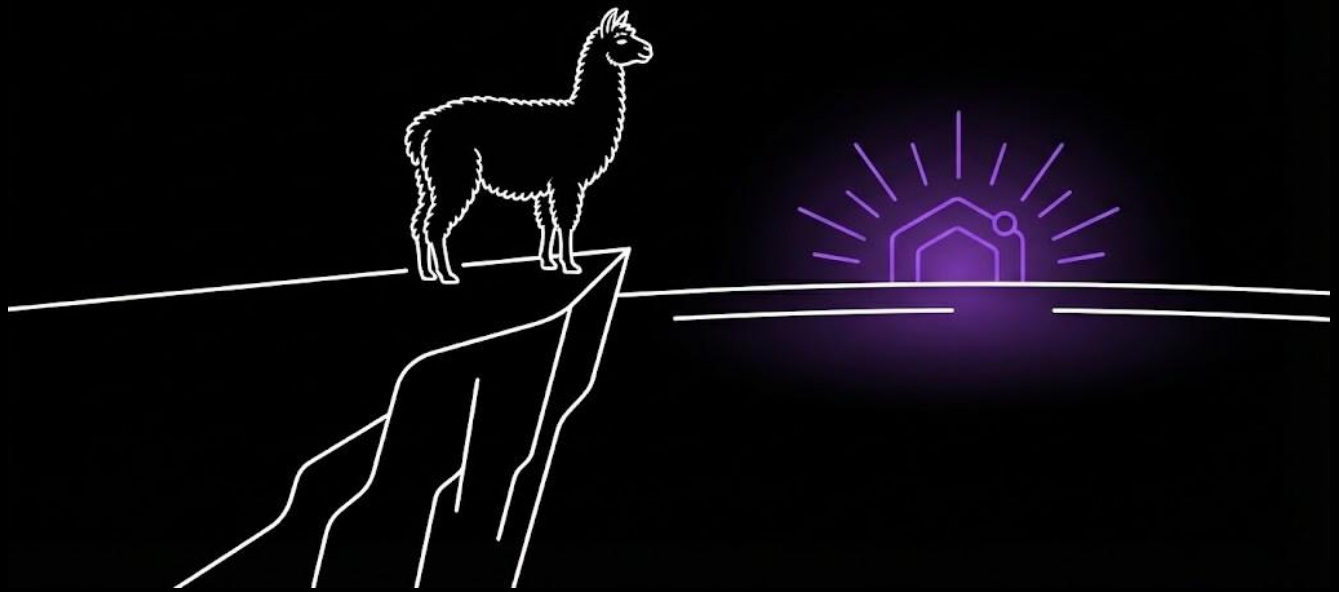
Blockchain, not Bitcoin

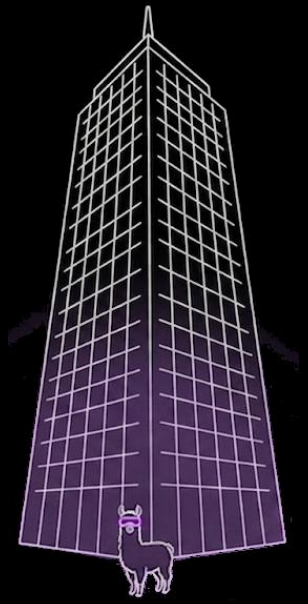
Hardware optimization
- Unused GPUs
- Fewer new mines?

Energy optimization
- Where it's cheaper
- Green energy surplus



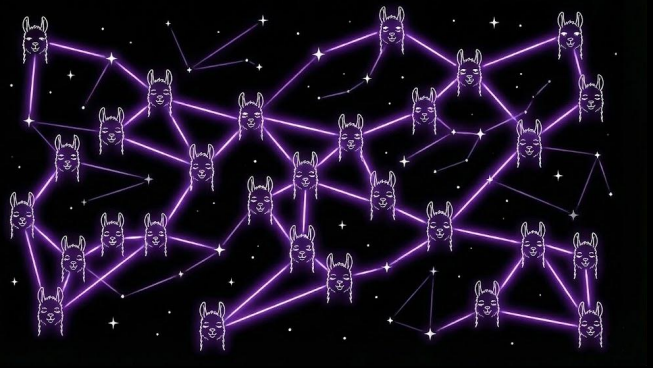
4. Epilogue





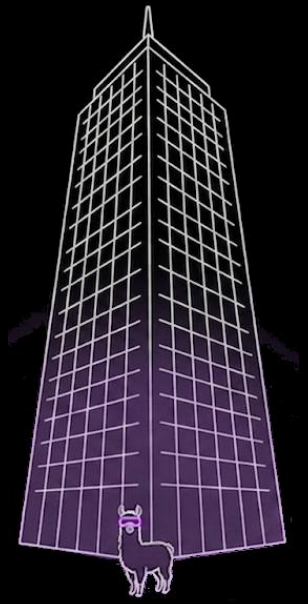
vs.





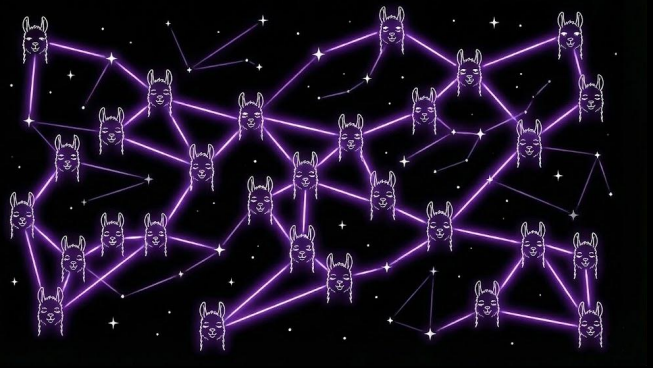
Monolith
(Cloud Oligopoly)

Distributed Architectures
(Open Stacks, Protocols, Decentralization)



with





Monolith
(Cloud)

Distributed Architectures
(Open Stacks, Protocols, Decentralization)

Next Stack - Physical AI ?



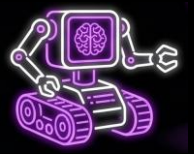
Physical Agents

From “Chat” in the browser to "Act" in the world



Mobile Revolution

Personalized AI (Edge Training, Privacy)



Open Robotics

Local and real-time for a moving robots



TinyML & Ambient Intelligence

Battery-powered, "always-on" open models for IoT

Open, Local, and Distributed is necessary for **AI to have a body**


FLASH
FORWARD
20??


FLASH
FORWARD
2036



the-human-protocol.com

OPEN COMMUNITY FOR AI

WORLDLINE



raphiki.github.io